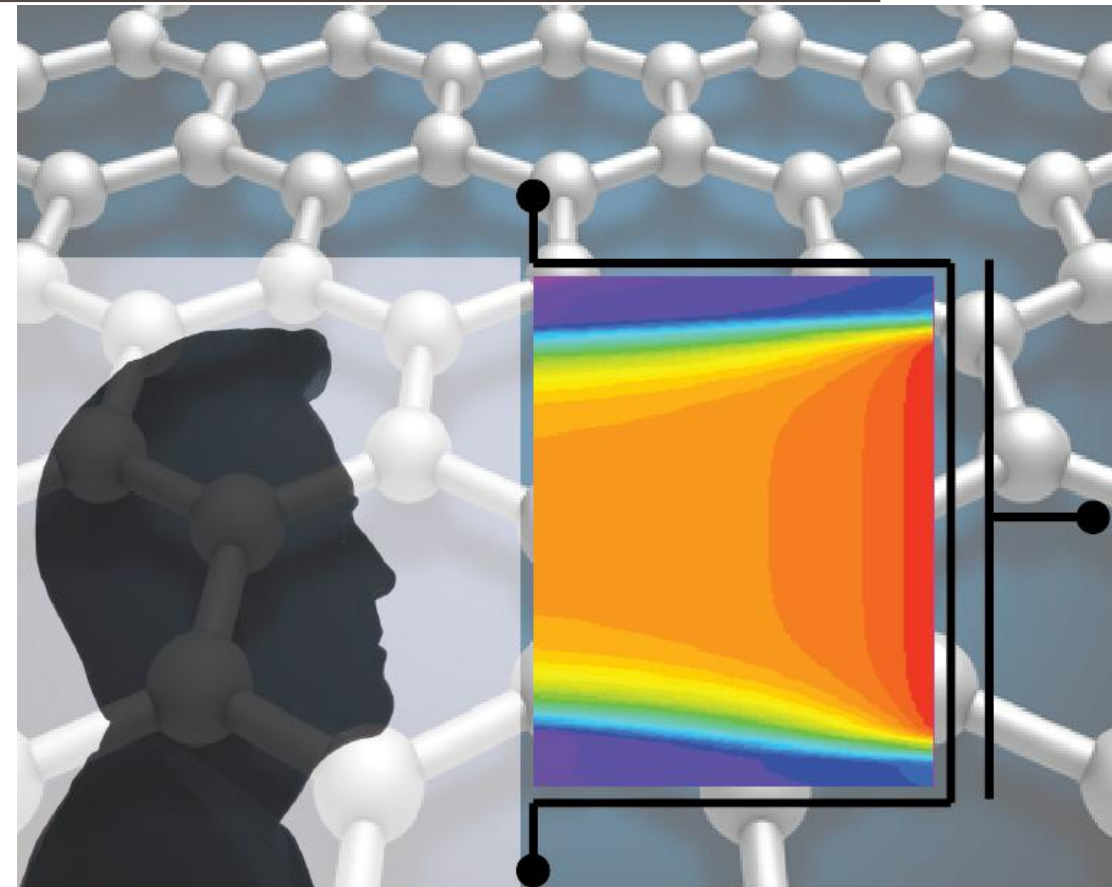


# ADVANCED SOLID-STATE DEVICES

Week-10, Lecture-1

Sneh Saurabh  
24<sup>th</sup> March, 2025



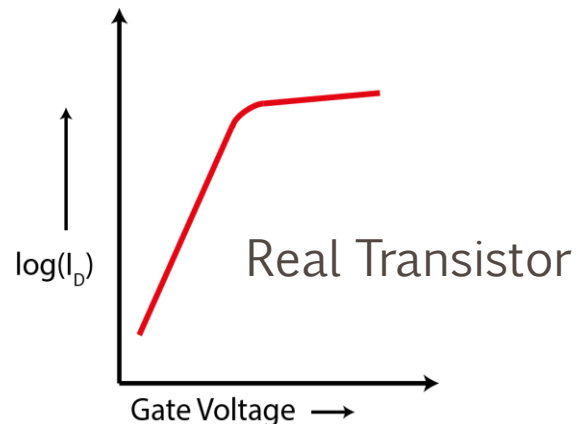
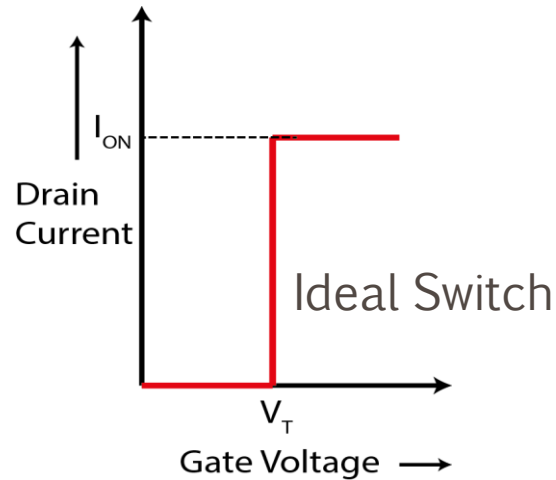
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# MOSFET: Non-Ideal Effects



## Subthreshold Current

# Subthreshold Behavior: Measure



- Transistors are employed as a switch in digital circuits
- An ideal switch has an *abrupt transition* from the OFF-state state to the ON-state

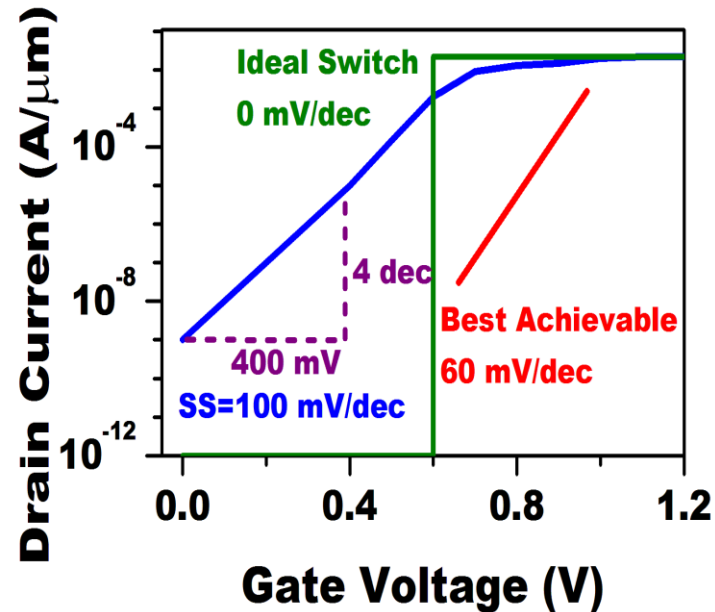
It is desirable that transistors behave as an ideal switch

- Can operate at very low supply voltages: How?
- Low Power consumption

- Realistic transistors are not ideal switch
- MOSFETs transitions from OFF-state to ON-state exponentially

- How to quantify how good a transistor is compared to an ideal switch?
- Measure *subthreshold swing*

# Subthreshold Behavior: Subthreshold Swing

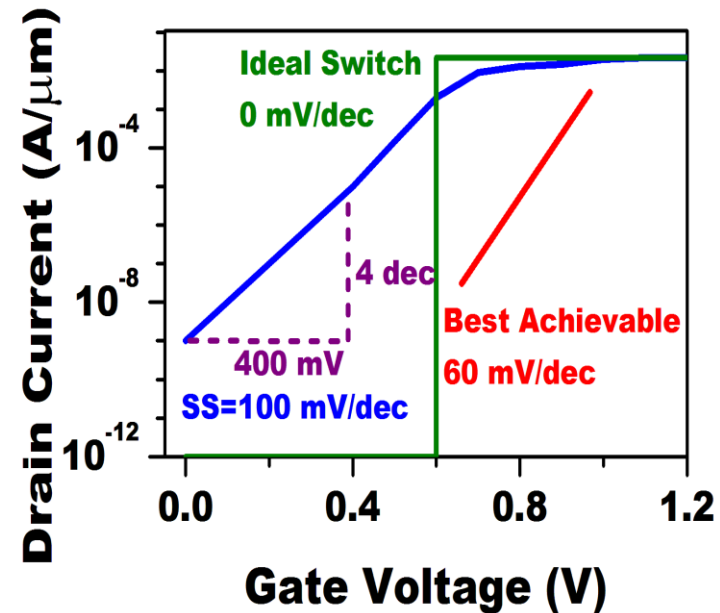


Is smaller subthreshold swing desirable for operation as a switch or larger subthreshold swing?

- Subthreshold swing measures the abruptness of the transition from the OFF-state to the ON-state
- **Definition:** Gate voltage required to increase the *drain current by 10 times* [one decade, one order of magnitude] when operating in the subthreshold region
- What is the unit of subthreshold swing?
  - mV/decade

- Ideally subthreshold swing should be 0 mV/decade
- In designing transistor, one of the major objectives is to have subthreshold swing as small as possible

# Subthreshold Swing of Ideal MOSFET



The subthreshold current can be given as:

$$I_D \propto \left[ \exp\left(\frac{V_{GS}}{kT/q}\right) \right] \left[ 1 - \exp\left(-\frac{V_{DS}}{kT/q}\right) \right]$$

Ideal subthreshold swing:

$$SS_{ideal} = \ln(10) \cdot \frac{kT}{q}$$

At room temperature:

$$SS_{ideal} = 60 \text{ mV/decade}$$

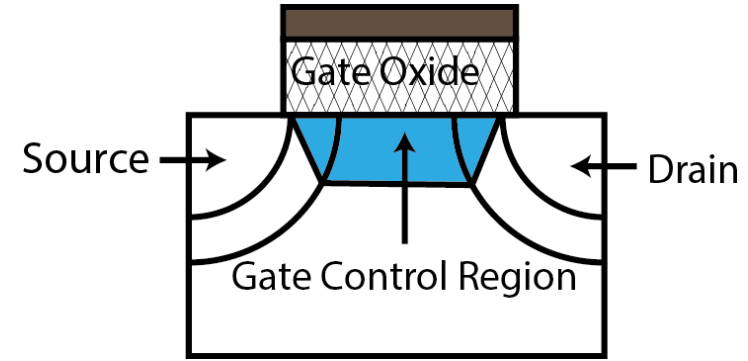
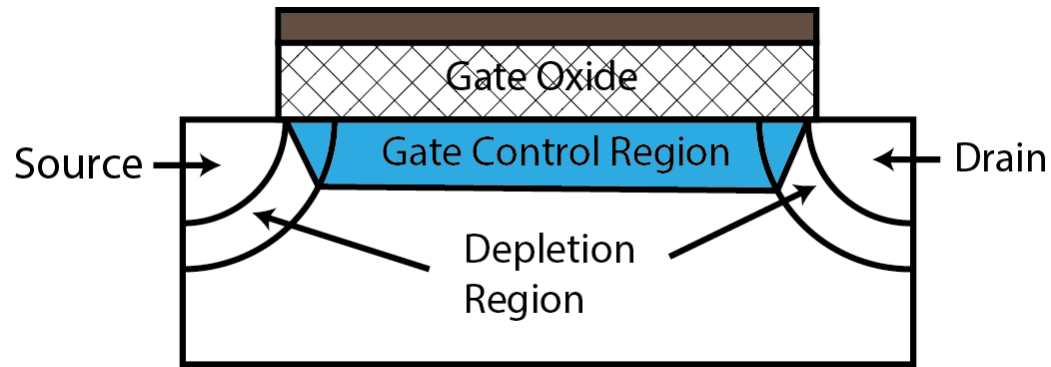
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# MOSFET: Non-Ideal Effects



## Drain-induced Barrier Lowering

# Drain Voltage Effects on Channel Potential



- *Ideally*. Potential energy in the channel should be controlled **only** by the gate voltage
- *In realistic devices*. Potential energy in the channel not only depends on the gate voltage but *also depends on the drain voltage*
- The *drain control* becomes prominent as the distance between the drain and the source i.e. the channel length is reduced
- This is a manifestation of *short-channel effects*

# Drain-induced Barrier Lowering (DIBL)

---

## For ideal MOSFET

- We had computed threshold voltage by assuming that the gate controls total channel charge (and assumed charge neutrality condition)

## For short-channel MOSFET

- The drain—channel and source—channel depletion regions reduce the fraction of channel charge directly controlled by the gate voltage
- Threshold voltage of the MOSFET reduced due to drain potential
- This phenomenon of lowering of barrier due to drain voltage is known as *Drain Induced Barrier Lowering (DIBL)*

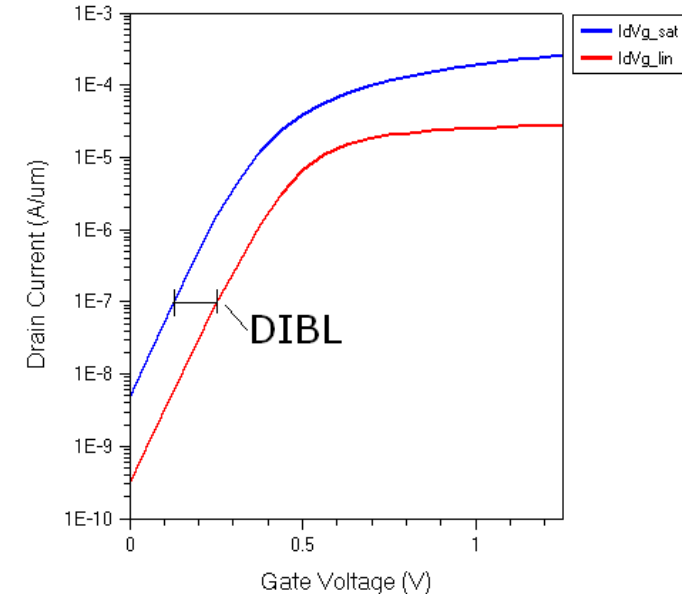
- DIBL is manifested in two forms:
  1. Reduction in threshold voltage at higher **drain voltages**
  2. Reduction in threshold voltage as the **channel length** is reduced



# DIBL: Impact of Drain Voltage on Threshold Voltage

- In ideal MOSFET, threshold voltage does not depend on the drain voltage

- In short n-channel MOSFET, when we apply higher  $V_{DS}$ , the depletion region around the drain widens
- As a result, effective channel length reduces
- Hence, the associated fraction of charge directly controlled by the gate decreases

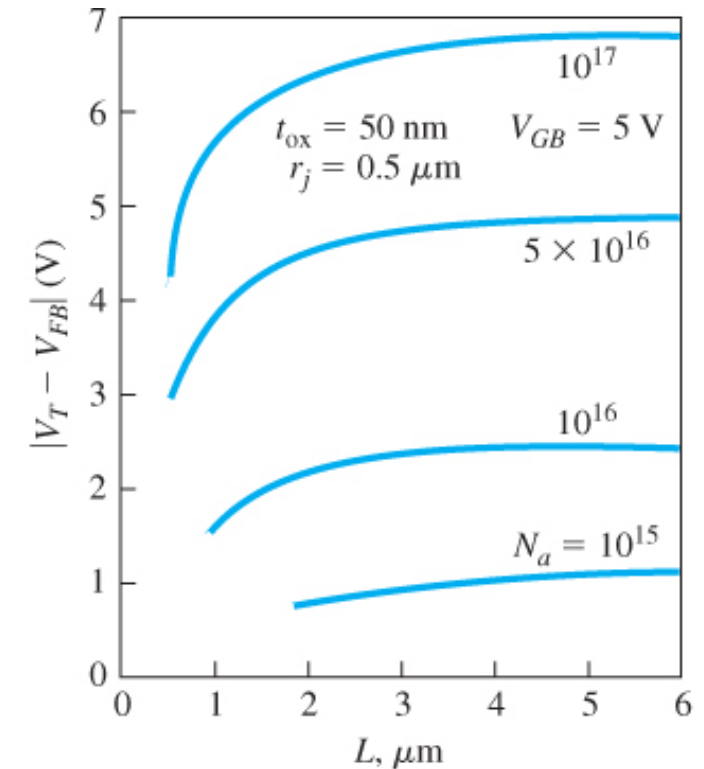


- For MOS with silicon p-type:  $V_T = \phi_{ms} + 2\phi_{Fp} + \frac{Q'_{s,max} - Q'_{ss}}{C'_{ox}}$

- For short channel and higher  $V_{DS}$ , effective  $Q'_{s,max}$  decreases, and hence threshold voltage decrease

# DIBL: Impact of Channel Length on Threshold Voltage

- In an ideal MOSFET, threshold voltage does not depend on the channel length
- With the decrease in channel length, the drain—channel and source—channel depletion regions reduce the fraction of channel charge  $Q'_{s,max}$  directly controlled by the gate voltage
  - Results in threshold voltage decrease with channel length
- Threshold voltage decreases with channel length when length becomes shorter than  $2\ \mu\text{m}$
- DIBL pose serious challenges in *scaling of transistors*
- Reduction in threshold voltage at smaller channel length further increases OFF-state current



**Figure 11.16** | Threshold voltage versus channel length for various substrate dopings. (From Yau [26].)

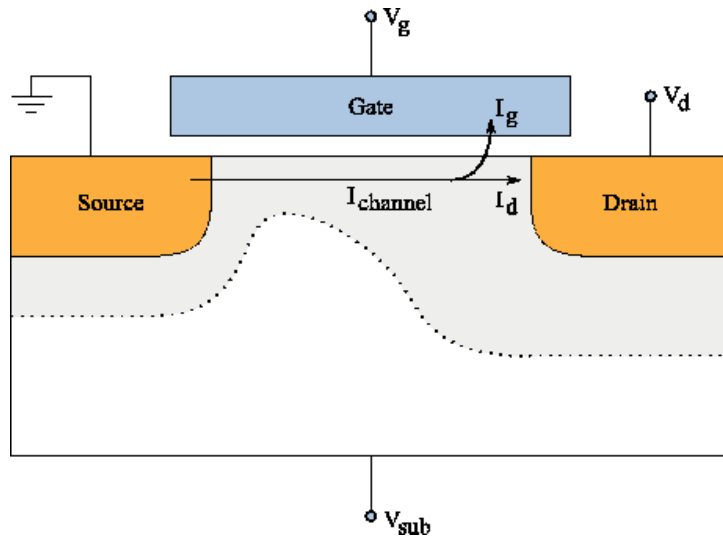
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# MOSFET: Non-Ideal Effects



## Hot Electron Effects

# Hot Electron Effect: Basics



## *Hot electron effect:*

- When high electric field is present, electrons attain very high kinetic energy
- Electrons with sufficient kinetic energy can tunnel through gate oxide barrier and contribute to gate current or get trapped inside the gate oxide
- Trapped electrons in gate oxide shift the threshold voltage of the transistor and creates *reliability* problems

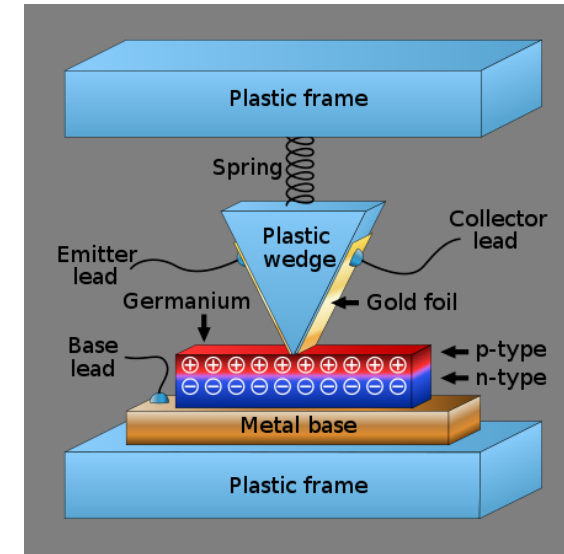
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# Bipolar Junction Transistor (BJT)

# BJT: Introduction

- The first solid-state transistor was **bipolar point-contact transistor** in December 1947
  - This could amplify signals
  - Later, **bipolar junction transistor** was invented in December 1948
- BJT remained device of choice for ICs for three decades (till 1980)
    - MOSFET was difficult to fabricate because of interface states
- Now, CMOS technology is more popular
  - Still BJTs are preferred in several applications due to high speed, high gain, low output impedance

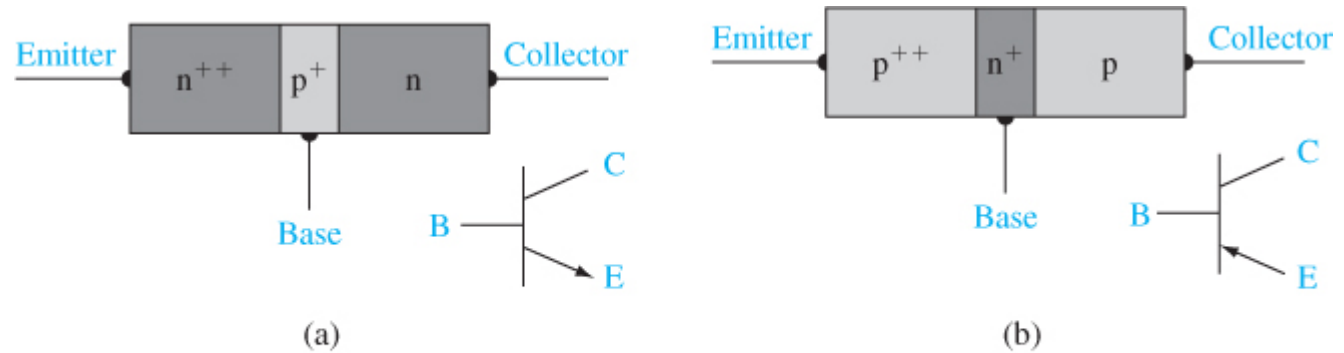


- Analog and mixed signal circuit: bandgap reference, amplifiers (high-frequency amplifiers, low-noise RF amplifiers), silicon bandgap temperature sensors, etc.
- Bipolar CMOS (BiCMOS) technology
- High voltage switches

# BJT: Basics (1)

- BJT consists of two PN junctions

- One of the regions is common for these two junctions
  - NP and PN: P region is shared, and we obtain **NPN transistor**
  - PN and NP: N is shared, and we get **PNP transistor**

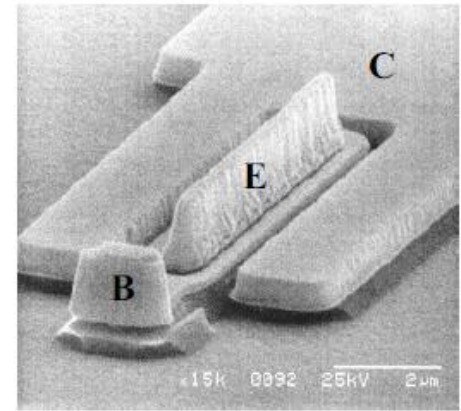


**Figure 12.1** | Simplified block diagrams and circuit symbols of (a) npn and (b) pnp bipolar transistors.

- These two junctions are interacting
  - The middle region is very thin
  - Interaction leads to *transistor action*
- The operation and characteristics of BJTs can be derived using the theory of PN junctions

## BJT: Basics (2)

- Unlike MOSFETs, BJTs are non-symmetric devices
- BJTs have three terminals:
  1. Emitter (E) : highly doped ( $\sim 10^{19} \text{cm}^{-3}$ )
  2. Base (B) : moderately doped region in between ( $\sim 10^{17} \text{cm}^{-3}$ )
  3. Collector (C): lightly doped ( $\sim 10^{15} \text{cm}^{-3}$ )





# References

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- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.
- Taur, Yuan, and Tak H. Ning. *Fundamentals of modern VLSI devices*. Cambridge university press, 2021.
- D. A. Neaman, "*Semiconductor Physics and Devices*"